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DVOC (Level-5)
1st Sem/ Trade DVOC (Auto, Servicing, AMT)
Subject : Modern Electric & Hybrid Vehicles

Time : 2 Hrs. M.M. : 50

SECTION-A

Note: Very Short answer type questions. All questions are compulsory (10x1=10)

Q.1 Hybrid vehicle.

Q.2 Unburned Hydrocarbons.

Q.3 AC Electrification System

Q.4 Fuel Cell Electric Vehicle.

Q.5 Power Flow control

Q.6 Drive controllers.

Q.7 DC Motor (Brushed).

Q.8 Power Convertors.

Q.9 Piezoelectric materials.

Q.10 Energy harvesting.

SECTION-B

Note: Short answer type questions. Attempt any six questions out of eight questions. (6x5=30)

Q.11 Write the Advantages of Hybrid Vehicles.

Q.12 Explain the basic concept of electric and hybrid traction.

Q.13 Explain power flow control in electric and hybrid electric vehicles.

Q.14 What are the basic requirements for electric motor?

Q.15 Why regenerative braking system is required? Explain.

Q.16 Explain the working principle of DC motor.

Q.17 What is regenerative braking and why is it more efficient?

Q.18 Explain in brief about the use of shock absorbers as vibration energy harvesters.

SECTION-C

Note: Long answer type questions. Attempt any one questions out of two questions. (1x10=10)

Q.19 Explain in detail electric and hybrid electric drive trains.

Q.20 Explain power flow control in electric and hybrid electric vehicles.